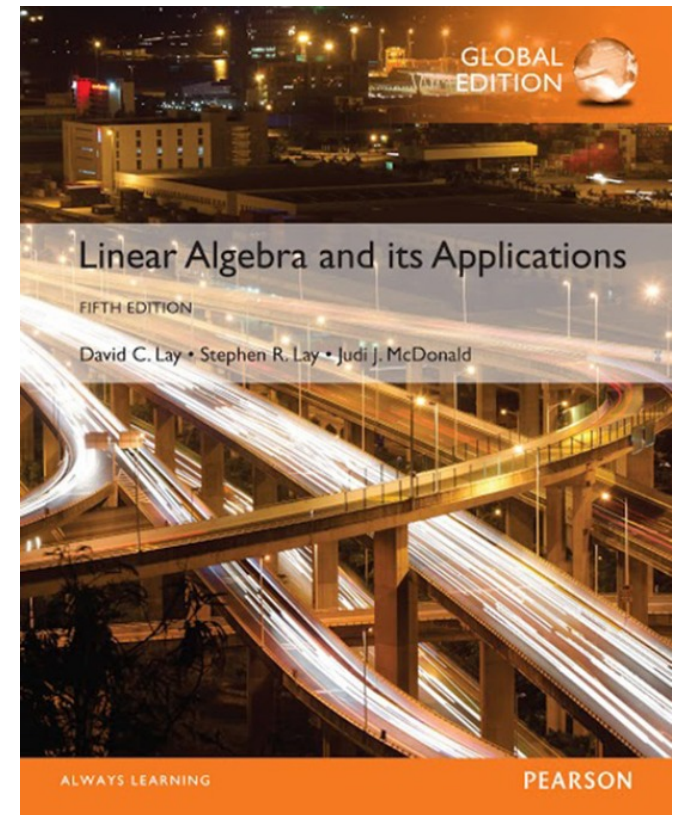


6

Orthogonality and Least Squares

6.1

INNER PRODUCT, LENGTH, AND ORTHOGONALITY



INNER PRODUCT

- If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, then we regard $\mathbf{u}$ and $\mathbf{v}$ as $n \times 1$ matrices.
- The transpose $\mathbf{u}^T$ is a $1 \times n$ matrix, and the matrix product $\mathbf{u}^T \mathbf{v}$ is a 1×1 matrix, which we write as a single real number (a scalar) without brackets.
- The number $\mathbf{u}^T \mathbf{v}$ is called the **inner product** of $\mathbf{u}$ and $\mathbf{v}$, and it is written as $u \cdot v$.
- This inner product is also referred to as a **dot product**.

INNER PRODUCT

- If $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$,

then the inner product of $\mathbf{u}$ and $\mathbf{v}$ is

$$\begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

INNER PRODUCT

- **Theorem 1:** Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in $\mathbb{R}^n$, and let c be a scalar. Then

a. $u \cdot v = v \cdot u$

b. $(u + v) \cdot w = u \cdot w + v \cdot w$

c. $(cu) \cdot v = c(u \cdot v) = u \cdot (cv)$

d. $u \cdot u \geq 0$, and $u \cdot u = 0$ if and only if $u = 0$

- Properties (b) and (c) can be combined several times to produce the following useful rule:

$$(c_1 u_1 + \dots + c_p u_p) \cdot w = c_1 (u_1 \cdot w) + \dots + c_p (u_p \cdot w)$$

THE LENGTH OF A VECTOR

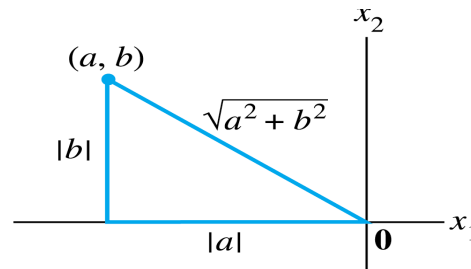
- If $\mathbf{v}$ is in $\mathbb{R}^n$, with entries $v_1, \dots, v_n$, then the square root of $\mathbf{v} \cdot \mathbf{v}$ is defined because $\mathbf{v} \cdot \mathbf{v}$ is nonnegative.
- **Definition:** The **length** (or **norm**) of $\mathbf{v}$ is the nonnegative scalar $\|\mathbf{v}\|$ defined by

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}, \text{ and } \|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$$

- Suppose $\mathbf{v}$ is in $\mathbb{R}^2$, say, $\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$

THE LENGTH OF A VECTOR

- If we identify $\mathbf{v}$ with a geometric point in the plane, as usual, then $\|\mathbf{v}\|$ coincides with the standard notion of the length of the line segment from the origin to $\mathbf{v}$.
- This follows from the Pythagorean Theorem applied to a triangle such as the one shown in the following figure.



Interpretation of $\|\mathbf{v}\|$ as length.

- For any scalar c , the length $c\mathbf{v}$ is $|c|$ times the length of $\mathbf{v}$. That is,
- $$\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$$

THE LENGTH OF A VECTOR

- A vector whose length is 1 is called a **unit vector**.
- If we *divide* a nonzero vector $\mathbf{v}$ by its length—that is, multiply by $1 / \|\mathbf{v}\|$ —we obtain a unit vector $\mathbf{u}$ because the length of $\mathbf{u}$ is $(1 / \|\mathbf{v}\|)\|\mathbf{v}\|$.
- The process of creating $\mathbf{u}$ from $\mathbf{v}$ is sometimes called **normalizing** $\mathbf{v}$, and we say that $\mathbf{u}$ is *in the same direction* as $\mathbf{v}$.

THE LENGTH OF A VECTOR

- **Example 2:** Let $\mathbf{v} = (1, -2, 2, 0)$. Find a unit vector $\mathbf{u}$ in the same direction as $\mathbf{v}$.
- **Solution:** First, compute the length of $\mathbf{v}$:

$$\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v} = (1)^2 + (-2)^2 + (2)^2 + (0)^2 = 9$$

$$\|\mathbf{v}\| = \sqrt{9} = 3$$

- Then, multiply $\mathbf{v}$ by $1 / \|\mathbf{v}\|$ to obtain

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|} \mathbf{v} = \frac{1}{3} \mathbf{v} = \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \\ 0 \end{bmatrix}$$

DISTANCE IN $\mathbb{R}^n$

- To check that $\|\mathbf{u}\| = 1$, it suffices to show that $\|\mathbf{u}\|^2 = 1$.

$$\begin{aligned}\|\mathbf{u}\|^2 &= \mathbf{u} \cdot \mathbf{u} = \left(\frac{1}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + (0)^2 \\ &= \frac{1}{9} + \frac{4}{9} + \frac{4}{9} + 0 = 1\end{aligned}$$

- **Definition:** For $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$, the **distance between $\mathbf{u}$ and $\mathbf{v}$** , written as $\text{dist}(\mathbf{u}, \mathbf{v})$, is the length of the vector $\mathbf{u} - \mathbf{v}$. That is,

$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

DISTANCE IN $\mathbb{R}^n$

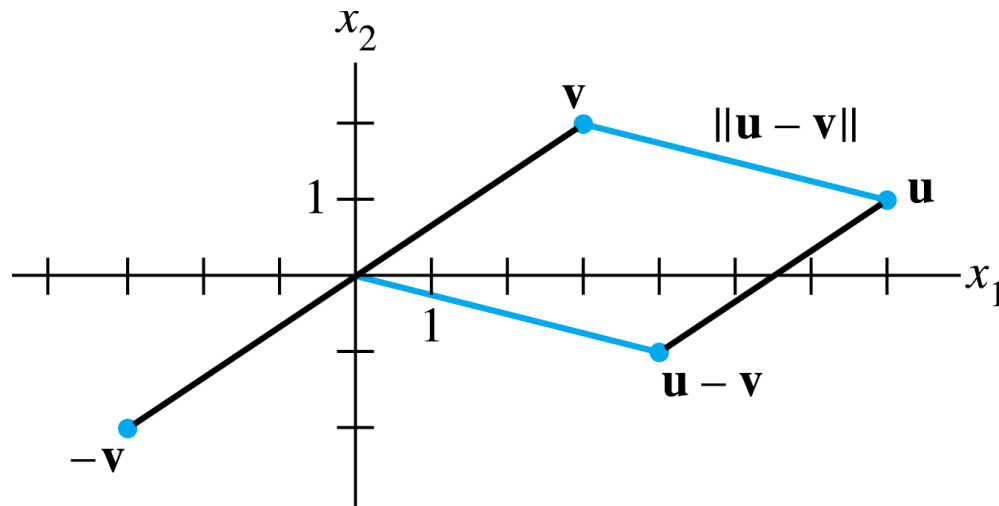
- **Example 4:** Compute the distance between the vectors $\mathbf{u} = (7, 1)$ and $\mathbf{v} = (3, 2)$.
- **Solution:** Calculate

$$\mathbf{u} - \mathbf{v} = \begin{bmatrix} 7 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

$$\|\mathbf{u} - \mathbf{v}\| = \sqrt{4^2 + (-1)^2} = \sqrt{17}$$

- The vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} - \mathbf{v}$ are shown in the figure on the next slide.
- When the vector $\mathbf{u} - \mathbf{v}$ is added to $\mathbf{v}$, the result is $\mathbf{u}$.

DISTANCE IN $\mathbb{R}^n$

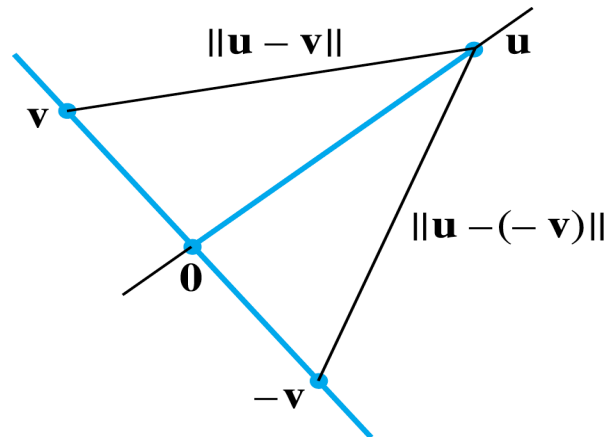


The distance between $\mathbf{u}$ and $\mathbf{v}$ is the length of $\mathbf{u} - \mathbf{v}$.

- Notice that the parallelogram in the above figure shows that the distance from $\mathbf{u}$ to $\mathbf{v}$ is the same as the distance from $\mathbf{u} - \mathbf{v}$ to $\mathbf{0}$.

ORTHOGONAL VECTORS

- Consider $\mathbb{R}^2$ or $\mathbb{R}^3$ and two lines through the origin determined by vectors $\mathbf{u}$ and $\mathbf{v}$.
- See the figure below. The two lines shown in the figure are geometrically perpendicular if and only if the distance from $\mathbf{u}$ to $\mathbf{v}$ is the same as the distance from $\mathbf{u}$ to $-\mathbf{v}$.



- This is the same as requiring the squares of the distances to be the same.

ORTHOGONAL VECTORS

- Now

$$\begin{aligned}[dist(u, -v)]^2 &= \|u - (-v)\|^2 = \|u + v\|^2 \\&= (u + v) \cdot (u + v) \\&= u \cdot (u + v) + v \cdot (u + v) \\&= u \cdot u + u \cdot v + v \cdot u + v \cdot v \\&= \|u\|^2 + \|v\|^2 + 2u \cdot v\end{aligned}$$

Theorem 1(b)

Theorem 1(a), (b)

Theorem 1(a)

- The same calculations with v and $-v$ interchanged show that

$$\begin{aligned}[dist(u, v)]^2 &= \|u\|^2 + \|-v\|^2 + 2u \cdot (-v) \\&= \|u\|^2 + \|v\|^2 - 2u \cdot v\end{aligned}$$

ORTHOGONAL VECTORS

- The two squared distances are equal if and only if $2u \cdot v = -2u \cdot v$, which happens if and only if $u \cdot v = 0$.
- This calculation shows that when vectors $\mathbf{u}$ and $\mathbf{v}$ are identified with geometric points, the corresponding lines through the points and the origin are perpendicular if and only if $u \cdot v = 0$.
- **Definition:** Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are **orthogonal** (to each other) if $u \cdot v = 0$.
- The zero vector is orthogonal to every vector in $\mathbb{R}^n$ because $0^T \mathbf{v} = 0$ for all $\mathbf{v}$.

THE PYTHOGOREAN THEOREM

- **Theorem 2:** Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Orthogonal Complements

- If a vector $\mathbf{z}$ is orthogonal to every vector in a subspace W of $\mathbb{R}^n$, then $\mathbf{z}$ is said to be **orthogonal to W** .
- The set of all vectors $\mathbf{z}$ that are orthogonal to W is called the **orthogonal complement** of W and is denoted by $W^\perp$ (and read as “ W perpendicular” or simply “ W perp”).

ORTHOGONAL COMPLEMENTS

1. A vector $\mathbf{x}$ is in $W^\perp$ if and only if $\mathbf{x}$ is orthogonal to every vector in a set that spans W .
2. $W^\perp$ is a subspace of $\mathbb{R}^n$.

- **Theorem 3:** Let A be an $m \times n$ matrix. The orthogonal complement of the row space of A is the null space of A , and the orthogonal complement of the column space of A is the null space of A^T :

$$\begin{array}{ccc} & \text{and} & \\ (\text{Row } A)^\perp = \text{Nul } A & & (\text{Col } A)^\perp = \text{Nul } A^T \end{array}$$

ORTHOGONAL COMPLEMENTS

- **Proof:** The row-column rule for computing $A\mathbf{x}$ shows that if $\mathbf{x}$ is in $\text{Nul}A$, then $\mathbf{x}$ is orthogonal to each row of A (with the rows treated as vectors in $\mathbb{R}^n$).
- Since the rows of A span the row space, $\mathbf{x}$ is orthogonal to $\text{Row } A$.
- Conversely, if $\mathbf{x}$ is orthogonal to $\text{Row } A$, then $\mathbf{x}$ is certainly orthogonal to each row of A , and hence $A\mathbf{x} = \mathbf{0}$.
- This proves the first statement of the theorem.

ORTHOGONAL COMPLEMENTS

- Since this statement is true for any matrix, it is true for A^T .
- That is, the orthogonal complement of the row space of A^T is the null space of A^T .
- This proves the second statement, because $\text{Row } A^T = \text{Col } A$.

ANGLES IN $\mathbb{R}^2$ AND $\mathbb{R}^3$ (OPTIONAL)

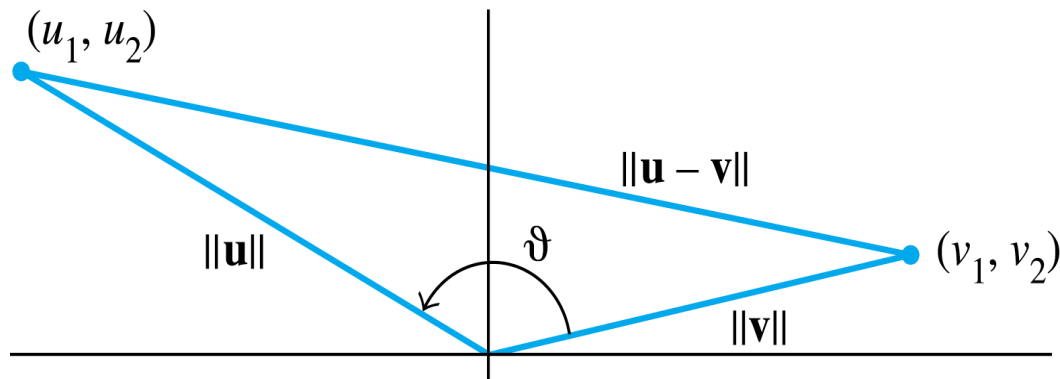
- If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors in either $\mathbb{R}^2$ or $\mathbb{R}^3$, then there is a nice connection between their inner product and the angle $\mathcal{G}$ between the two line segments from the origin to the points identified with $\mathbf{u}$ and $\mathbf{v}$.

- The formula is

$$(2) \quad u \cdot v = \|u\| \|v\| \cos \mathcal{G}$$

- To verify this formula for vectors in $\mathbb{R}^2$, consider the triangle shown in the figure on the next slide with sides of lengths, $\|u\|$, $\|v\|$, and $\|u - v\|$.

ANGLES IN $\mathbb{R}^2$ AND $\mathbb{R}^3$ (OPTIONAL)



The angle between two vectors.

- By the law of cosines,

$$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\cos \vartheta$$

which can be rearranged to produce the equations on the next slide.

ANGLES IN $\mathbb{R}^2$ AND $\mathbb{R}^3$ (OPTIONAL)

$$\begin{aligned}\|\mathbf{u}\| \|\mathbf{v}\| \cos \mathcal{G} &= \frac{1}{2} \left[\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 \right] \\ &= \frac{1}{2} \left[u_1^2 + u_2^2 + v_1^2 + v_2^2 - (u_1 - v_1)^2 - (u_2 - v_2)^2 \right] \\ &= u_1 v_1 + u_2 v_2 \\ &= \mathbf{u} \cdot \mathbf{v}\end{aligned}$$

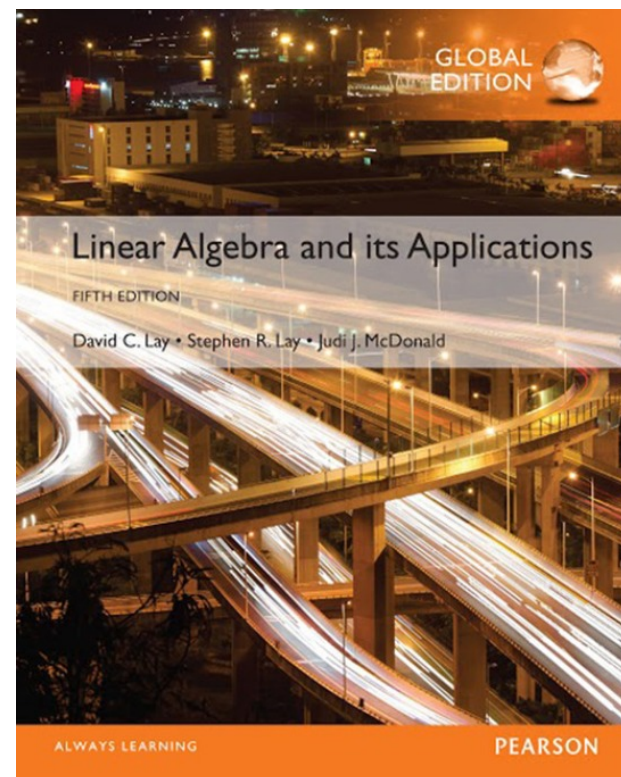
- The verification for $\mathbb{R}^3$ is similar.
- When $n > 3$, formula (2) may be used to *define* the angle between two vectors in $\mathbb{R}^n$.
- In statistics, the value of $\cos \mathcal{G}$ defined by (2) for suitable vectors $\mathbf{u}$ and $\mathbf{v}$ is called a *correlation coefficient*.

6

Orthogonality and Least Squares

6.2

ORTHOGONAL SETS



ORTHOGONAL SETS

- A set of vectors $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ in $\mathbb{R}^n$ is said to be an **orthogonal set** if each pair of distinct vectors from the set is orthogonal, that is, if $u_i \cdot u_j = 0$ whenever $i \neq j$.
- **Theorem 4:** If $S = \{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S .

ORTHOGONAL SETS

- **Proof:** If $0 = c_1 \mathbf{u}_1 + \cdots + c_p \mathbf{u}_p$ for some scalars $c_1, \dots, c_p$, then

$$\begin{aligned} 0 &= 0 \cdot \mathbf{u}_1 = (c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + \cdots + c_p \mathbf{u}_p) \cdot \mathbf{u}_1 \\ &= (c_1 \mathbf{u}_1) \cdot \mathbf{u}_1 + (c_2 \mathbf{u}_2) \cdot \mathbf{u}_1 + \cdots + (c_p \mathbf{u}_p) \cdot \mathbf{u}_1 \\ &= c_1 (\mathbf{u}_1 \cdot \mathbf{u}_1) + c_2 (\mathbf{u}_2 \cdot \mathbf{u}_1) + \cdots + c_p (\mathbf{u}_p \cdot \mathbf{u}_1) \\ &= c_1 (\mathbf{u}_1 \cdot \mathbf{u}_1) \end{aligned}$$

because $\mathbf{u}_1$ is orthogonal to $\mathbf{u}_2, \dots, \mathbf{u}_p$.

- Since $\mathbf{u}_1$ is nonzero, $\mathbf{u}_1 \cdot \mathbf{u}_1$ is not zero and so $c_1 = 0$
- Similarly, $c_2, \dots, c_p$ must be zero.

ORTHOGONAL SETS

- Thus S is linearly independent.
- **Definition:** An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.
- **Theorem 5:** Let $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y}$ in W , the weights in the linear combination

$$\mathbf{y} = c_1 \mathbf{u}_1 + \cdots + c_p \mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j} \quad (j = 1, \dots, p)$$

ORTHOGONAL SETS

- **Proof:** The orthogonality of $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ shows that

$$y \cdot u_1 = (c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + \dots + c_p \mathbf{u}_p) \cdot u_1 = c_1 u_1 \cdot u_1$$

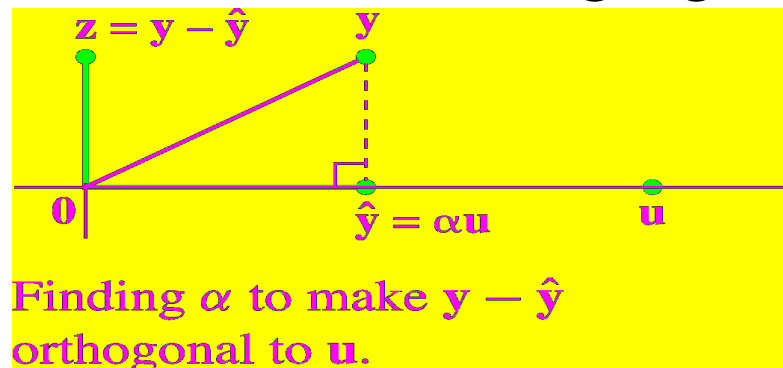
- Since $u_1 \cdot u_1$ is not zero, the equation above can be solved for c_1 .
- To find c_j for $j = 2, \dots, p$, compute $y \cdot u_j$ and solve for c_j .

AN ORTHOGONAL PROJECTION

- Given a nonzero vector $\mathbf{u}$ in $\mathbb{R}^n$, consider the problem of decomposing a vector $\mathbf{y}$ in $\mathbb{R}^n$ into the sum of two vectors, one a multiple of $\mathbf{u}$ and the other orthogonal to $\mathbf{u}$.
- We wish to write

$$(1) \quad \mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

where $\hat{\mathbf{y}} = \alpha \mathbf{u}$ for some scalar α and $\mathbf{z}$ is some vector orthogonal to $\mathbf{u}$. See the following figure.



AN ORTHOGONAL PROJECTION

- Given any scalar α , let $\mathbf{z} = \mathbf{y} - \alpha\mathbf{u}$, so that (1) is satisfied.
- Then $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to $\mathbf{u}$ if and only if
$$0 = (\mathbf{y} - \alpha\mathbf{u}) \cdot \mathbf{u} = \mathbf{y} \cdot \mathbf{u} - (\alpha\mathbf{u}) \cdot \mathbf{u} = \mathbf{y} \cdot \mathbf{u} - \alpha(\mathbf{u} \cdot \mathbf{u})$$
- That is, (1) is satisfied with $\mathbf{z}$ orthogonal to $\mathbf{u}$ if and

only if $\alpha = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}$ and $\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$.

- The vector $\hat{\mathbf{y}}$ is called the **orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$** , and the vector $\mathbf{z}$ is called the **component of $\mathbf{y}$ orthogonal to $\mathbf{u}$** .

AN ORTHOGONAL PROJECTION

- If c is any nonzero scalar and if $\mathbf{u}$ is replaced by $c\mathbf{u}$ in the definition of $\hat{\mathbf{y}}$, then the orthogonal projection of $\mathbf{y}$ onto $c\mathbf{u}$ is exactly the same as the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$.
- Hence this projection is determined by the *subspace* L spanned by $\mathbf{u}$ (the line through $\mathbf{u}$ and $\mathbf{0}$).
- Sometimes $\hat{\mathbf{y}}$ is denoted by $\text{proj}_L \mathbf{y}$ and is called the **orthogonal projection of $\mathbf{y}$ onto L** .
- That is,

$$\hat{\mathbf{y}} = \text{proj}_L \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} \quad (2)$$

AN ORTHOGONAL PROJECTION

- **Example 3:** Let $\mathbf{y} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$. Find the

orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$. Then write $\mathbf{y}$ as the sum of two orthogonal vectors, one in $\text{Span}\{\mathbf{u}\}$ and one orthogonal to $\mathbf{u}$.

- **Solution:** Compute

$$\mathbf{y} \cdot \mathbf{u} = \begin{bmatrix} 7 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 40$$

$$\mathbf{u} \cdot \mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 20$$

AN ORTHOGONAL PROJECTION

- The orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$ is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = \frac{40}{20} \mathbf{u} = 2 \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}$$

and the component of $\mathbf{y}$ orthogonal to $\mathbf{u}$ is

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 7 \\ 6 \end{bmatrix} - \begin{bmatrix} 8 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

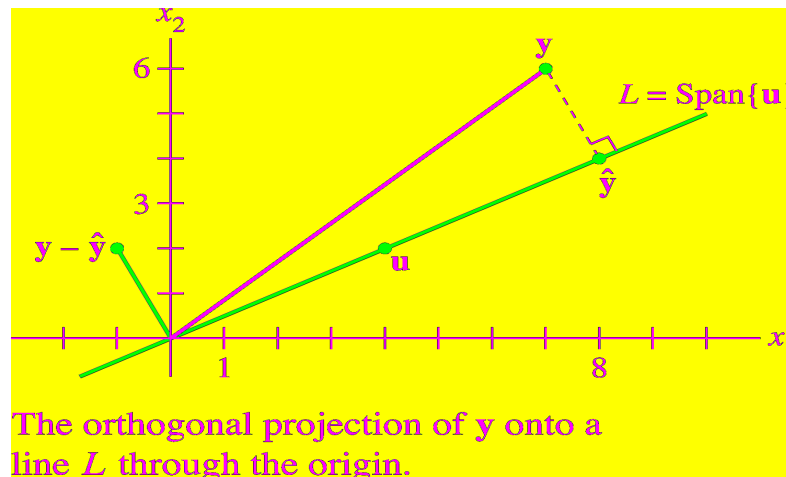
- The sum of these two vectors is $\mathbf{y}$.

AN ORTHOGONAL PROJECTION

- That is,

$$\begin{array}{c} \begin{bmatrix} 7 \\ 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} \\ \uparrow \qquad \qquad \uparrow \qquad \qquad \uparrow \\ y \qquad \qquad \hat{y} \qquad \qquad (y - \hat{y}) \end{array}$$

- The decomposition of y is illustrated in the following figure:



AN ORTHOGONAL PROJECTION

- *Note:* If the calculations above are correct, then $\{\hat{y}, y - \hat{y}\}$ will be an orthogonal set.

- As a check, compute

$$\hat{y} \cdot (y - \hat{y}) = \begin{bmatrix} 8 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = -8 + 8 = 0$$

- Since the line segment in the figure on the previous slide between y and $\hat{y}$ is perpendicular to L , by construction of $\hat{y}$, the point identified with $\hat{y}$ is the closest point of L to y .

ORTHONORMAL SETS

- A set $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an **orthonormal set** if it is an orthogonal set of unit vectors.
- If W is the subspace spanned by such a set, then $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an **orthonormal basis** for W , since the set is automatically linearly independent, by Theorem 4.
- The simplest example of an orthonormal set is the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ for $\mathbb{R}^n$.
- Any nonempty subset of $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is orthonormal, too.

ORTHONORMAL SETS

- **Example 2:** Show that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthonormal basis of $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 3 / \sqrt{11} \\ 1 / \sqrt{11} \\ 1 / \sqrt{11} \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 / \sqrt{6} \\ 2 / \sqrt{6} \\ 1 / \sqrt{6} \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 / \sqrt{66} \\ -4 / \sqrt{66} \\ 7 / \sqrt{66} \end{bmatrix}$$

- **Solution:** Compute

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = -3 / \sqrt{66} + 2 / \sqrt{66} + 1 / \sqrt{66} = 0$$

$$\mathbf{v}_1 \cdot \mathbf{v}_3 = -3 / \sqrt{726} - 4 / \sqrt{726} + 7 / \sqrt{726} = 0$$

ORTHONORMAL SETS

$$\mathbf{v}_2 \cdot \mathbf{v}_3 = 1 / \sqrt{396} - 8 / \sqrt{396} + 7 / \sqrt{396} = 0$$

- Thus $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthogonal set.

- Also, $\mathbf{v}_1 \cdot \mathbf{v}_1 = 9 / 11 + 1 / 11 + 1 / 11 = 1$

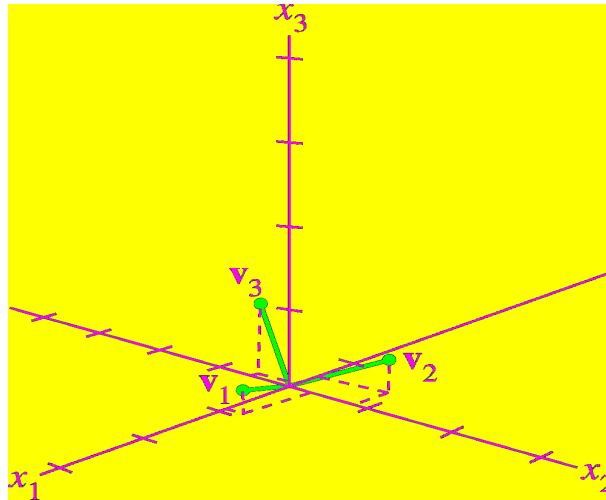
$$\mathbf{v}_2 \cdot \mathbf{v}_2 = 1 / 6 + 4 / 6 + 1 / 6 = 1$$

$$\mathbf{v}_3 \cdot \mathbf{v}_3 = 1 / 66 + 16 / 66 + 49 / 66 = 1$$

which shows that $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ are unit vectors.

- Thus $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthonormal set.
- Since the set is linearly independent, its three vectors form a basis for $\mathbb{R}^3$. See the figure on the next slide.

ORTHONORMAL SETS



- When the vectors in an orthogonal set of nonzero vectors are *normalized* to have unit length, the new vectors will still be orthogonal, and hence the new set will be an orthonormal set.

ORTHONORMAL SETS

- **Theorem 6:** An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I$.
- **Proof:** To simplify notation, we suppose that U has only three columns, each a vector in $\mathbb{R}^m$.
- Let $U = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{bmatrix}$ and compute

$$U^T U = \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \mathbf{u}_3^T \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1^T \mathbf{u}_1 & \mathbf{u}_1^T \mathbf{u}_2 & \mathbf{u}_1^T \mathbf{u}_3 \\ \mathbf{u}_2^T \mathbf{u}_1 & \mathbf{u}_2^T \mathbf{u}_2 & \mathbf{u}_2^T \mathbf{u}_3 \\ \mathbf{u}_3^T \mathbf{u}_1 & \mathbf{u}_3^T \mathbf{u}_2 & \mathbf{u}_3^T \mathbf{u}_3 \end{bmatrix} \quad (4)$$

ORTHONORMAL SETS

- The entries in the matrix at the right are inner products, using transpose notation.
- The columns of U are orthogonal if and only if
$$\mathbf{u}_1^T \mathbf{u}_2 = \mathbf{u}_2^T \mathbf{u}_1 = 0, \mathbf{u}_1^T \mathbf{u}_3 = \mathbf{u}_3^T \mathbf{u}_1 = 0, \mathbf{u}_2^T \mathbf{u}_3 = \mathbf{u}_3^T \mathbf{u}_2 = 0 \quad (5)$$
- The columns of U all have unit length if and only if
$$\mathbf{u}_1^T \mathbf{u}_1 = 1, \mathbf{u}_2^T \mathbf{u}_2 = 1, \mathbf{u}_3^T \mathbf{u}_3 = 1 \quad (6)$$
- The theorem follows immediately from (4)–(6).

ORTHONORMAL SETS

- **Theorem 7:** Let U be an $m \times n$ matrix with orthonormal columns, and let $\mathbf{x}$ and $\mathbf{y}$ be in $\mathbb{R}^n$.

Then

$$\|U\mathbf{x}\| = \|\mathbf{x}\|$$

$$(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$$

$$(U\mathbf{x}) \cdot (U\mathbf{y}) = 0 \text{ if and only if } \mathbf{x} \cdot \mathbf{y} = 0$$

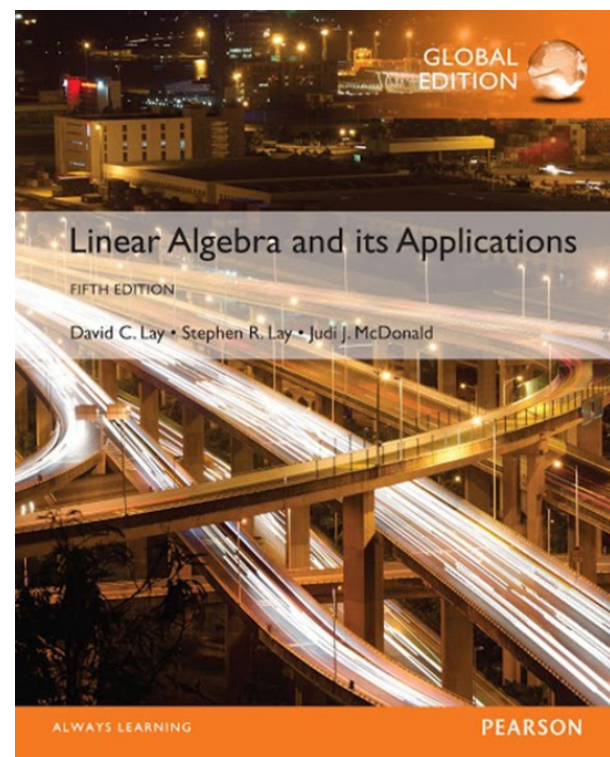
- Properties (a) and (c) say that the linear mapping $\mathbf{x} \mapsto U\mathbf{x}$ preserves lengths and orthogonality.

6

Orthogonality and Least Squares

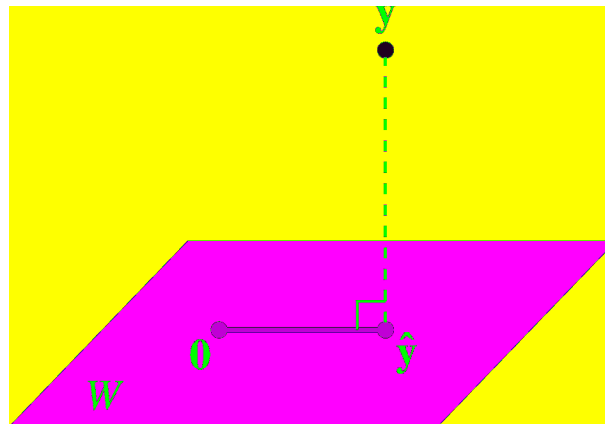
6.3

ORTHOGONAL PROJECTIONS



ORTHOGONAL PROJECTIONS

- The orthogonal projection of a point in $\mathbb{R}^2$ onto a line through the origin has an important analogue in $\mathbb{R}^n$.
- Given a vector $\mathbf{y}$ and a subspace W in $\mathbb{R}^n$, there is a vector $\hat{\mathbf{y}}$ in W such that (1) $\hat{\mathbf{y}}$ is the unique vector in W for which $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to W , and (2) $\hat{\mathbf{y}}$ is the unique vector in W closest to $\mathbf{y}$. See the following figure.



THE ORTHOGONAL DECOMPOSITION THEOREM

- These two properties of $\hat{y}$ provide the key to finding the least-squares solutions of linear systems.
- **Theorem 8:** Let W be a subspace of $\mathbb{R}^n$. Then each y in $\mathbb{R}^n$ can be written uniquely in the form

$$(1) \quad y = \hat{y} + z$$

where $\hat{y}$ is in W and z is in $W^\perp$.

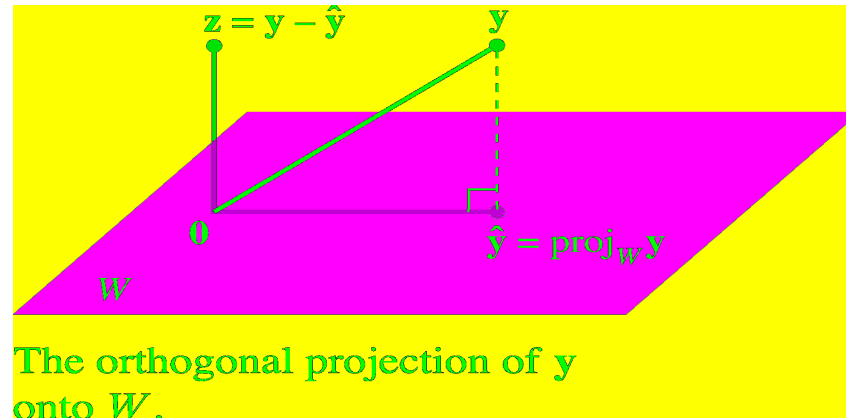
- In fact, if $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is any orthogonal basis of W , then

$$(2) \quad \hat{y} = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \dots + \frac{y \cdot u_p}{u_p \cdot u_p} u_p$$

and $z = y - \hat{y}$.

THE ORTHOGONAL DECOMPOSITION THEOREM

- The vector $\hat{y}$ in (1) is called the **orthogonal projection of y onto W** and often is written as $\text{proj}_W y$. See the following figure:



- **Proof:** Let $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ be any orthogonal basis for W , and define $\hat{y}$ by (2).
- Then $\hat{y}$ is in W because $\hat{y}$ is a linear combination of the basis $\mathbf{u}_1, \dots, \mathbf{u}_p$.

THE ORTHOGONAL DECOMPOSITION THEOREM

- Let $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.
- Since $\mathbf{u}_1$ is orthogonal to $\mathbf{u}_2, \dots, \mathbf{u}_p$, it follows from (2) that

$$\begin{aligned} \mathbf{z} \cdot \mathbf{u}_1 &= (\mathbf{y} - \hat{\mathbf{y}}) \cdot \mathbf{u}_1 = \mathbf{y} \cdot \mathbf{u}_1 - \left(\frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \right) \mathbf{u}_1 \cdot \mathbf{u}_1 - 0 \dots 0 - 0 \\ &= \mathbf{y} \cdot \mathbf{u}_1 - \mathbf{y} \cdot \mathbf{u}_1 = 0 \end{aligned}$$

- Thus $\mathbf{z}$ is orthogonal to $\mathbf{u}_1$.
- Similarly, $\mathbf{z}$ is orthogonal to each $\mathbf{u}_j$ in the basis for W .
- Hence $\mathbf{z}$ is orthogonal to every vector in W .
- That is, $\mathbf{z}$ is in $W^\perp$.

THE ORTHOGONAL DECOMPOSITION THEOREM

- To show that the decomposition in (1) is unique, suppose $\mathbf{y}$ can also be written as $\mathbf{y} = \hat{\mathbf{y}}_1 + \mathbf{z}_1$, with $\hat{\mathbf{y}}_1$ in W and $\mathbf{z}_1$ in $W^\perp$.

- Then $\hat{\mathbf{y}} + \mathbf{z} = \hat{\mathbf{y}}_1 + \mathbf{z}_1$ (since both sides equal $\mathbf{y}$), and so

$$\hat{\mathbf{y}} - \hat{\mathbf{y}}_1 = \mathbf{z}_1 - \mathbf{z}$$

- This equality shows that the vector $\mathbf{v} = \hat{\mathbf{y}} - \hat{\mathbf{y}}_1$ is in W and in $W^\perp$ (because $\mathbf{z}_1$ and $\mathbf{z}$ are both in $W^\perp$, and $W^\perp$ is a subspace).
- Hence $\mathbf{v} \cdot \mathbf{v} = 0$, which shows that $\mathbf{v} = \mathbf{0}$.
- This proves that $\hat{\mathbf{y}} = \hat{\mathbf{y}}_1$ and also $\mathbf{z}_1 = \mathbf{z}$.

THE ORTHOGONAL DECOMPOSITION THEOREM

- The uniqueness of the decomposition (1) shows that the orthogonal projection $\hat{y}$ depends only on W and not on the particular basis used in (2).
- **Example 1:** Let $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, and $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

Observe that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$. Write $\mathbf{y}$ as the sum of a vector in W and a vector orthogonal to W .

THE ORTHOGONAL DECOMPOSITION THEOREM

- **Solution:** The orthogonal projection of $\mathbf{y}$ onto W is

$$\begin{aligned}\hat{\mathbf{y}} &= \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 \\ &= \frac{9}{30} \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{3}{6} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \frac{9}{30} \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{15}{30} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix}\end{aligned}$$

- Also

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} = \begin{bmatrix} 7/5 \\ 0 \\ 14/5 \end{bmatrix}$$

THE ORTHOGONAL DECOMPOSITION THEOREM

- Theorem 8 ensures that $y - \hat{y}$ is in $W^\perp$.
- To check the calculations, verify that $y - \hat{y}$ is orthogonal to both $\mathbf{u}_1$ and $\mathbf{u}_2$ and hence to all of W .
- The desired decomposition of $\mathbf{y}$ is

$$\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} + \begin{bmatrix} 7/5 \\ 0 \\ 14/5 \end{bmatrix}$$

PROPERTIES OF ORTHOGONAL PROJECTIONS

- If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthogonal basis for W and if $\mathbf{y}$ happens to be in W , then the formula for $\text{proj}_W \mathbf{y}$ is exactly the same as the representation of $\mathbf{y}$ given in Theorem 5 in Section 6.2.
- In this case, $\text{proj}_W \mathbf{y} = \mathbf{y}$.
- If $\mathbf{y}$ is in $W = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$, then $\text{proj}_W \mathbf{y} = \mathbf{y}$.

THE BEST APPROXIMATION THEOREM

- **Theorem 9:** Let W be a subspace of $\mathbb{R}^n$, let $\mathbf{y}$ be any vector in $\mathbb{R}^n$, and let $\hat{\mathbf{y}}$ be the orthogonal projection of $\mathbf{y}$ onto W . Then $\hat{\mathbf{y}}$ is the closest point in W to $\mathbf{y}$, in the sense that

$$(3) \quad \|\mathbf{y} - \hat{\mathbf{y}}\| < \|\mathbf{y} - \mathbf{v}\|$$

for all $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$.

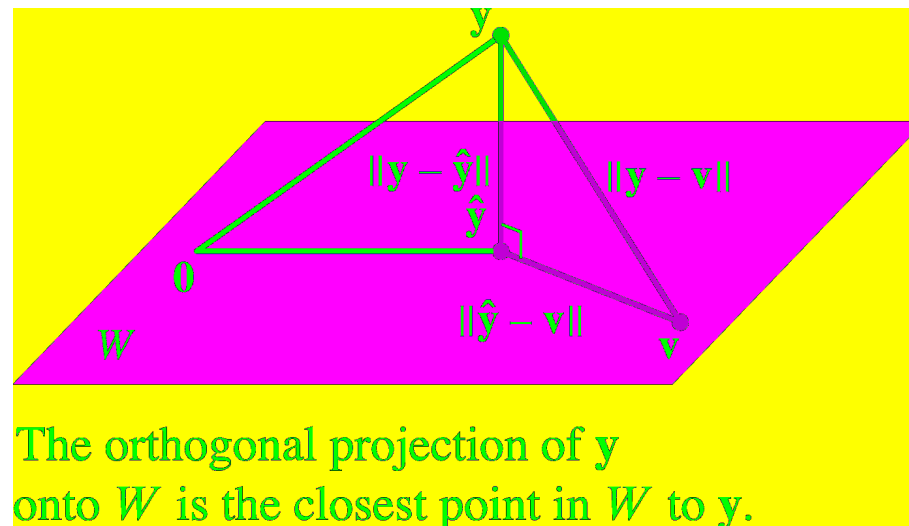
- The vector $\hat{\mathbf{y}}$ in Theorem 9 is called **the best approximation to $\mathbf{y}$ by elements of W** .
- The distance from $\mathbf{y}$ to $\mathbf{v}$, given by $\|\mathbf{y} - \mathbf{v}\|$, can be regarded as the “error” of using $\mathbf{v}$ in place of $\mathbf{y}$.
- Theorem 9 says that this error is minimized when $\mathbf{v} = \hat{\mathbf{y}}$.

THE BEST APPROXIMATION THEOREM

- Inequality (3) leads to a new proof that $\hat{\mathbf{y}}$ does not depend on the particular orthogonal basis used to compute it.
- If a different orthogonal basis for W were used to construct an orthogonal projection of $\mathbf{y}$, then this projection would also be the closest point in W to $\mathbf{y}$, namely, $\hat{\mathbf{y}}$.

THE BEST APPROXIMATION THEOREM

- **Proof:** Take $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$. See the following figure:



- Then $\hat{\mathbf{y}} - \mathbf{v}$ is in W .
- By the Orthogonal Decomposition Theorem, $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to W .
- In particular, $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to $\hat{\mathbf{y}} - \mathbf{v}$ (which is in W).

THE BEST APPROXIMATION THEOREM

- Since

$$y - v = (y - \hat{y}) + (\hat{y} - v)$$

the Pythagorean Theorem gives

$$\|y - v\|^2 = \|y - \hat{y}\|^2 + \|\hat{y} - v\|^2$$

- (See the colored right triangle in the figure on the previous slide. The length of each side is labeled.)
- Now $\|\hat{y} - v\|^2 > 0$ because $\hat{y} - v \neq 0$, and so inequality (3) follows immediately.

PROPERTIES OF ORTHOGONAL PROJECTIONS

- **Example 4:** The distance from a point $\mathbf{y}$ in $\mathbb{R}^n$ to a subspace W is defined as the distance from $\mathbf{y}$ to the nearest point in W . Find the distance from $\mathbf{y}$ to $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$, where

$$\mathbf{y} = \begin{bmatrix} -1 \\ -5 \\ 10 \end{bmatrix}, \mathbf{u}_1 = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

- **Solution:** By the Best Approximation Theorem, the distance from $\mathbf{y}$ to W is $\|\mathbf{y} - \hat{\mathbf{y}}\|$, where $\hat{\mathbf{y}} = \text{proj}_W \mathbf{y}$.

PROPERTIES OF ORTHOGONAL PROJECTIONS

- Since $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for W ,

$$\hat{\mathbf{y}} = \frac{15}{30}\mathbf{u}_1 + \frac{-21}{6}\mathbf{u}_2 = \frac{1}{2}\begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix} - \frac{7}{2}\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -8 \\ 4 \end{bmatrix}$$

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} -1 \\ -5 \\ 10 \end{bmatrix} - \begin{bmatrix} -1 \\ -8 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 6 \end{bmatrix}$$

$$\|\mathbf{y} - \hat{\mathbf{y}}\|^2 = 3^2 + 6^2 = 45$$

- The distance from $\mathbf{y}$ to W is $\sqrt{45} = 3\sqrt{5}$.

PROPERTIES OF ORTHOGONAL PROJECTIONS

- **Theorem 10:** If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthonormal basis for a subspace W of $\mathbb{R}^n$, then

$$\text{proj}_W \mathbf{y} = (\mathbf{y} \cdot \mathbf{u}_1)\mathbf{u}_1 + (\mathbf{y} \cdot \mathbf{u}_2)\mathbf{u}_2 + \cdots + (\mathbf{y} \cdot \mathbf{u}_p)\mathbf{u}_p \quad (4)$$

If $U = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_p \end{bmatrix}$, then

$$\text{for all } \mathbf{y} \text{ in } \mathbb{R}^n \quad \text{proj}_W \mathbf{y} = UU^T \mathbf{y} \quad (5)$$

- **Proof:** Formula (4) follows immediately from (2) in Theorem 8.

PROPERTIES OF ORTHOGONAL PROJECTIONS

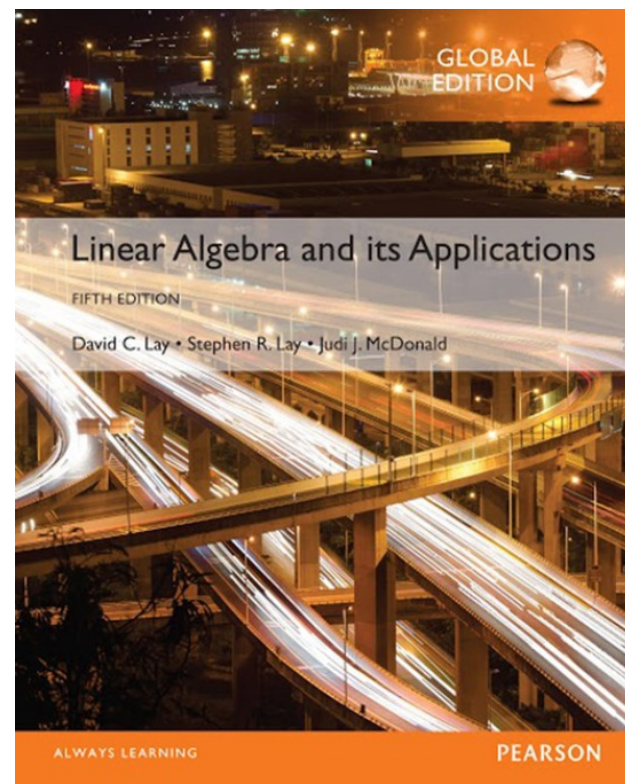
- Also, (4) shows that $\text{proj}_W \mathbf{y}$ is a linear combination of the columns of U using the weights $\mathbf{y} \cdot \mathbf{u}_1, \mathbf{y} \cdot \mathbf{u}_2, \dots, \mathbf{y} \cdot \mathbf{u}_p$.
- The weights can be written as $\mathbf{u}_1^T \mathbf{y}, \mathbf{u}_2^T \mathbf{y}, \dots, \mathbf{u}_p^T \mathbf{y}$, showing that they are the entries in $U^T \mathbf{y}$ and justifying (5).

6

Orthogonality and Least Squares

6.4

THE GRAM-SCHMIDT PROCESS



THE GRAM-SCHMIDT PROCESS

- **Theorem 11: The Gram-Schmidt Process**

- Given a basis $\{x_1, \dots, x_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define

$$\begin{aligned} v_1 &= x_1 \\ v_2 &= x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1 \\ v_3 &= x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2 \\ &\vdots \\ v_p &= x_p - \frac{x_p \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_p \cdot v_2}{v_2 \cdot v_2} v_2 - \dots - \frac{x_p \cdot v_{p-1}}{v_{p-1} \cdot v_{p-1}} v_{p-1} \end{aligned}$$

- Then $\{v_1, \dots, v_p\}$ is an orthogonal basis for W . In addition

$$\text{Span}\{v_1, \dots, v_k\} = \text{Span}\{x_1, \dots, x_k\} \quad \text{for } 1 \leq k \leq p \quad (1)$$

THE GRAM-SCHMIDT PROCESS

- **Proof** For $1 \leq k \leq p$, let $W_k = \text{Span}\{x_1, \dots, x_k\}$. Set $v_1 = x_1$, so that $\text{Span}\{v_1\} = \text{Span}\{x_1\}$. Suppose, for some $k < p$, we have constructed $v_1, \dots, v_k$ so that $\{v_1, \dots, v_k\}$ is an orthogonal basis for W_k . Define

$$v_{k+1} = x_{k+1} - \text{proj}_{W_k} x_{k+1} \quad (2)$$

- By the Orthogonal Decomposition Theorem, v_{k+1} is orthogonal to W_k . Furthermore, $v_{k+1} \neq 0$ because x_{k+1} is not in $W_k = \text{Span}\{x_1, \dots, x_k\}$
- Hence $\{v_1, \dots, v_{k+1}\}$ is an orthogonal set of nonzero vectors in the $(k+1)$ -dimensional space W_{k+1} . By the Basis Theorem in Section 4.5, this set is an orthogonal basis for W_{k+1} . Hence $W_{k+1} = \text{Span}\{v_1, \dots, v_{k+1}\}$. When $k+1 = p$, the process stops.

ORTHONORMAL BASES

- **Example 3** Example 1 constructed the orthogonal basis

$$v_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

- An orthonormal basis is

$$u_1 = \frac{1}{\|v_1\|} v_1 = \frac{1}{\sqrt{45}} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix}$$

$$u_2 = \frac{1}{\|v_2\|} v_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

QR FACTORIZATION OF MATRICES

- **Theorem 12: The QR Factorization**
- If A is an $m \times n$ matrix with linearly independent columns, then A can be factored as $A = QR$, where Q is an $m \times n$ matrix whose columns form an orthonormal basis for $\text{Col } A$ and R is an $n \times n$ upper triangular invertible matrix with positive entries on its diagonal.
- **Proof** The columns of A form a basis $\{x_1, \dots, x_n\}$ for $\text{Col } A$. Construct an orthonormal basis $\{u_1, \dots, u_n\}$ for $W = \text{Col } A$ with property (1) in Theorem 11. This basis may be constructed by the Gram-Schmidt process or some other means.

QR FACTORIZATION OF MATRICES

- Let

$$Q = [u_1 \ u_2 \ \dots \ u_n]$$

- For $k = 1, \dots, n$, x_k is in $\text{Span}\{x_1, \dots, x_k\} = \text{Span}\{u_1, \dots, u_k\}$. So there are constants, $r_{1k}, \dots, r_{kk}$, such that

$$x_k = r_{1k}u_1 + \dots + r_{kk}u_k + 0 \cdot u_{k+1} + \dots + 0 \cdot u_n$$

- We may assume that $r_{kk} > 0$. This shows that x_k is a linear combination of the columns of Q using as weights the entries in the vector

QR FACTORIZATION OF MATRICES

$$r_k = \begin{bmatrix} r_{1k} \\ \cdot \\ \cdot \\ \cdot \\ r_{kk} \\ 0 \\ \cdot \\ \cdot \\ \cdot \\ 0 \end{bmatrix}$$

- That is, $x_k = Qr_k$ for $k = 1, \dots, n$. Let $R = [r_1 \dots r_n]$. Then
$$A = [x_1 \dots x_n] = [Qr_1 \dots Qr_n] = QR$$
- The fact that R is invertible follows easily from the fact that the columns of A are linearly independent.

QR FACTORIZATION OF MATRICES

- **Example 4** Find a QR factorization of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.
- **Solution** The columns of A are the vectors x_1 , x_2 , and x_3 in Example 2. An orthogonal basis for $\text{Col } A = \text{Span}\{x_1, x_2, x_3\}$ was found in that example:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ -2/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

QR FACTORIZATION OF MATRICES

- To simplify the arithmetic that follows, scale v_3 by letting $v_3 = 3v_3$. Then normalize the three vectors to obtain u_1 , u_2 , and u_3 , and use these vectors as the columns of Q :

$$Q = \begin{bmatrix} 1/2 & -3/\sqrt{12} & 0 \\ 1/2 & 1/\sqrt{12} & -2/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \end{bmatrix}.$$

- By construction, the first k columns of Q are an orthonormal basis of $\text{Span}\{x_1, \dots, x_k\}$.

QR FACTORIZATION OF MATRICES

- From the proof of Theorem 12, $A = QR$ for some R . To find R , observe that $Q^T Q = I$, because the columns of Q are orthonormal. Hence

$$Q^T A = Q^T (QR) = IR = R$$

- and

$$R = \begin{bmatrix} 1/2 & 1/2 & 1/2 & 1/2 \\ -3/\sqrt{12} & 1/\sqrt{12} & 1/\sqrt{12} & 1/\sqrt{12} \\ 0 & -2/\sqrt{6} & 1/\sqrt{12} & 1/\sqrt{6} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

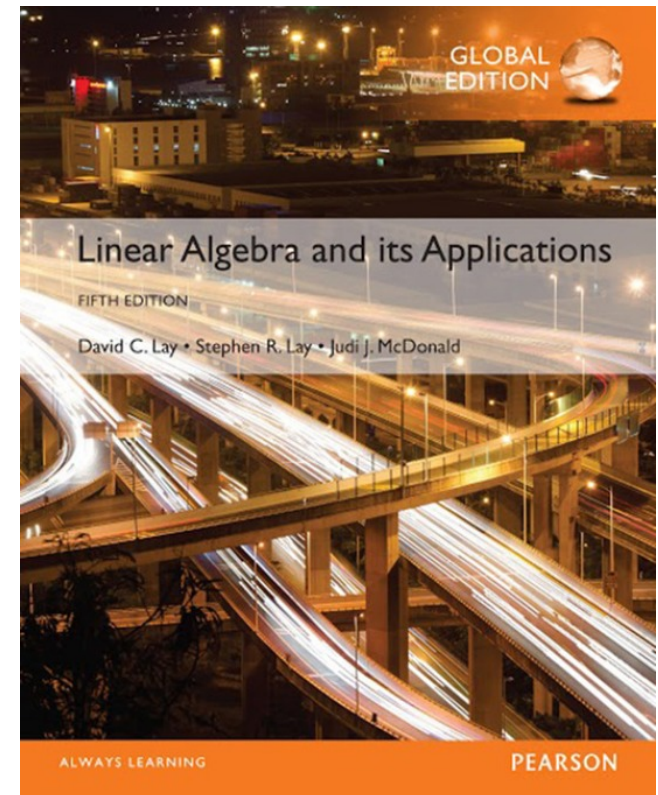
$$= \begin{bmatrix} 2 & 3/2 & 1 \\ 0 & 3/\sqrt{12} & 2/\sqrt{12} \\ 0 & 0 & 3/\sqrt{12} \end{bmatrix}$$

6

Orthogonality and Least Squares

6.5

LEAST-SQUARES PROBLEMS



LEAST-SQUARES PROBLEMS

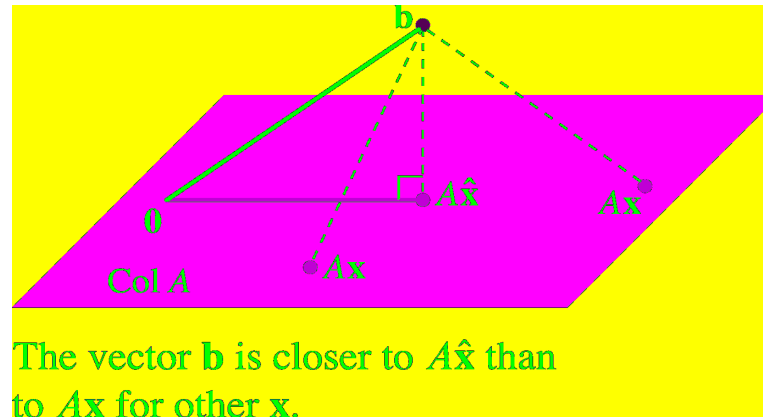
- **Definition:** If A is $m \times n$ and $\mathbf{b}$ is in $\mathbb{R}^m$, a **least-squares solution** of $A\mathbf{x} = \mathbf{b}$ is an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ such that

$$\|\mathbf{b} - A\hat{\mathbf{x}}\| \leq \|\mathbf{b} - A\mathbf{x}\|$$

for all $\mathbf{x}$ in $\mathbb{R}^n$.

- The most important aspect of the least-squares problem is that no matter what $\mathbf{x}$ we select, the vector $A\mathbf{x}$ will necessarily be in the column space, $\text{Col } A$.
- So we seek an $\mathbf{x}$ that makes $A\mathbf{x}$ the closest point in $\text{Col } A$ to $\mathbf{b}$. See the figure on the next slide.

LEAST-SQUARES PROBLEMS



- **Solution of the General Least-Squares Problem**

- Given A and $\mathbf{b}$, apply the Best Approximation Theorem to the subspace $\text{Col } A$.

- Let
$$\hat{\mathbf{b}} = \text{proj}_{\text{Col } A} \mathbf{b}$$

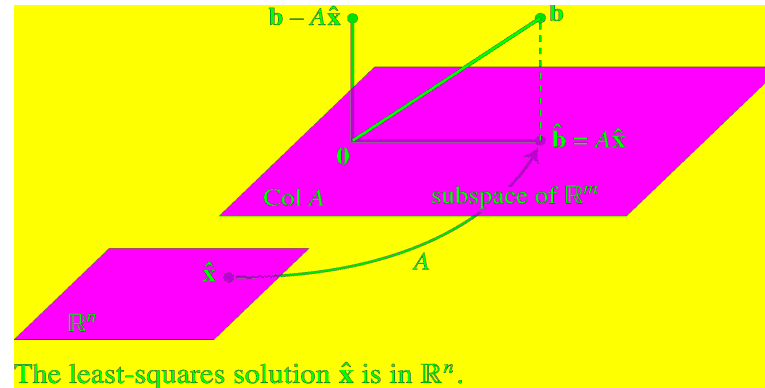
SOLUTION OF THE GENREAL LEAST-SQUARES PROBLEM

- Because $\hat{\mathbf{b}}$ is in the column space A , the equation $A\mathbf{x} = \hat{\mathbf{b}}$ is consistent, and there is an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ such that

$$(1) \quad A\hat{\mathbf{x}} = \hat{\mathbf{b}}$$

- Since $\hat{\mathbf{b}}$ is the closest point in $\text{Col } A$ to $\mathbf{b}$, a vector $\hat{\mathbf{x}}$ is a least-squares solution of $A\mathbf{x} = \mathbf{b}$ if and only if $\hat{\mathbf{x}}$ satisfies (1).
- Such an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ is a list of weights that will build $\hat{\mathbf{b}}$ out of the columns of A . See the figure on the next slide.

SOLUTION OF THE GENREAL LEAST-SQUARES PROBLEM



- Suppose $\hat{\mathbf{x}}$ satisfies $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$.
- By the Orthogonal Decomposition Theorem, the projection $\hat{\mathbf{b}}$ has the property that $\mathbf{b} - \hat{\mathbf{b}}$ is orthogonal to $\text{Col } A$, so $\mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to each column of A .
- If $\mathbf{a}_j$ is any column of A , then $\mathbf{a}_j \cdot (\mathbf{b} - A\hat{\mathbf{x}}) = 0$, and $\mathbf{a}_j^T (\mathbf{b} - A\hat{\mathbf{x}}) = 0$.

SOLUTION OF THE GENREAL LEAST-SQUARES PROBLEM

- Since each $\mathbf{a}_j^T$ is a row of A^T ,
$$A^T (\mathbf{b} - A\hat{\mathbf{x}}) = 0 \quad (2)$$

- Thus

$$A^T \mathbf{b} - A^T A \hat{\mathbf{x}} = 0$$

$$A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$$

- These calculations show that each least-squares solution of $A\mathbf{x} = \mathbf{b}$ satisfies the equation

$$A^T A \mathbf{x} = A^T \mathbf{b} \quad (3)$$

- The matrix equation (3) represents a system of equations called the **normal equations** for $A\mathbf{x} = \mathbf{b}$.
- A solution of (3) is often denoted by $\hat{\mathbf{x}}$.

SOLUTION OF THE GENREAL LEAST-SQUARES PROBLEM

- **Theorem 13:** The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ coincides with the nonempty set of solutions of the normal equation $A^T A\mathbf{x} = A^T \mathbf{b}$.
- **Proof:** The set of least-squares solutions is nonempty and each least-squares solution $\hat{\mathbf{x}}$ satisfies the normal equations.
- Conversely, suppose $\hat{\mathbf{x}}$ satisfies $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$.
- Then $\hat{\mathbf{x}}$ satisfies (2), which shows that $\mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to the rows of A^T and hence is orthogonal to the columns of A .
- Since the columns of A span $\text{Col } A$, the vector $\mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to all of $\text{Col } A$.

SOLUTION OF THE GENREAL LEAST-SQUARES PROBLEM

- Hence the equation

$$\mathbf{b} = A\hat{\mathbf{x}} + (\mathbf{b} - A\hat{\mathbf{x}})$$

is a decomposition of $\mathbf{b}$ into the sum of a vector in $\text{Col } A$ and a vector orthogonal to $\text{Col } A$.

- By the uniqueness of the orthogonal decomposition, $A\hat{\mathbf{x}}$ must be the orthogonal projection of $\mathbf{b}$ onto $\text{Col } A$.
- That is, $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$ and $\hat{\mathbf{x}}$ is a least-squares solution.

SOLUTION OF THE GENREAL LEAST-SQUARES PROBLEM

- **Example 1:** Find a least-squares solution of the inconsistent system $A\mathbf{x} = \mathbf{b}$ for

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$$

- **Solution:** To use normal equations (3), compute:

$$A^T A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$$

SOLUTION OF THE GENREAL LEAST-SQUARES PROBLEM

$$A^T \mathbf{b} = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

- Then the equation $A^T A \mathbf{x} = A^T \mathbf{b}$ becomes

$$\begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

SOLUTION OF THE GENREAL LEAST-SQUARES PROBLEM

- Row operations can be used to solve the system on the previous slide, but since $A^T A$ is invertible and 2×2 , it is probably faster to compute

$$(A^T A)^{-1} = \frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix}$$

and then solve $A^T A \mathbf{x} = A^T \mathbf{b}$ as

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

$$= \frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix} \begin{bmatrix} 19 \\ 11 \end{bmatrix} = \frac{1}{84} \begin{bmatrix} 84 \\ 168 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

SOLUTION OF THE GENREAL LEAST-SQUARES PROBLEM

- **Theorem 14:** Let A be an $m \times n$ matrix. The following statements are logically equivalent:
 - a. The equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution for each $\mathbf{b}$ in $\mathbb{R}^m$.
 - b. The columns of A are linearly independent.
 - c. The matrix $A^T A$ is invertible.

When these statements are true, the least-squares solution $\hat{\mathbf{x}}$ is given by

$$(4) \quad \hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

- When a least-squares solution $\hat{\mathbf{x}}$ is used to produce $A\hat{\mathbf{x}}$ as an approximation to $\mathbf{b}$, the distance from $\mathbf{b}$ to $A\hat{\mathbf{x}}$ is called the **least-squares error** of this approximation.

ALTERNATIVE CALCULATIONS OF LEAST-SQUARES SOLUTIONS

- **Example 4:** Find a least-squares solution of $A\mathbf{x} = \mathbf{b}$ for

$$A = \begin{bmatrix} 1 & -6 \\ 1 & -2 \\ 1 & 1 \\ 1 & 7 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 6 \end{bmatrix}$$

- **Solution:** Because the columns $\mathbf{a}_1$ and $\mathbf{a}_2$ of A are orthogonal, the orthogonal projection of $\mathbf{b}$ onto $\text{Col } A$ is given by

$$\hat{\mathbf{b}} = \frac{\mathbf{b} \cdot \mathbf{a}_1}{\mathbf{a}_1 \cdot \mathbf{a}_1} \mathbf{a}_1 + \frac{\mathbf{b} \cdot \mathbf{a}_2}{\mathbf{a}_2 \cdot \mathbf{a}_2} \mathbf{a}_2 = \frac{8}{4} \mathbf{a}_1 + \frac{45}{90} \mathbf{a}_2 \quad (5)$$

ALTERNATIVE CALCULATIONS OF LEAST-SQUARES SOLUTIONS

$$= \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} -3 \\ -1 \\ 1/2 \\ 7/2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 5/2 \\ 11/2 \end{bmatrix}$$

- Now that $\hat{\mathbf{b}}$ is known, we can solve $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$.
- But this is trivial, since we already know weights to place on the columns of A to produce $\hat{\mathbf{b}}$.
- It is clear from (5) that

$$\hat{\mathbf{x}} = \begin{bmatrix} 8/4 \\ 45/90 \end{bmatrix} = \begin{bmatrix} 2 \\ 1/2 \end{bmatrix}$$

ALTERNATIVE CALCULATIONS OF LEAST-SQUARES SOLUTIONS

- **Theorem 15:** Given an $m \times n$ matrix A with linearly independent columns, let $A = QR$ be a QR factorization of A . Then, for each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution, given by

$$(6) \quad \hat{\mathbf{x}} = R^{-1}Q^T\mathbf{b}$$

- **Proof:** Let $\hat{\mathbf{x}} = R^{-1}Q^T\mathbf{b}$.
- Then

$$A\hat{\mathbf{x}} = QR\hat{\mathbf{x}} = QR R^{-1}Q^T\mathbf{b} = QQ^T\mathbf{b}$$

ALTERNATIVE CALCULATIONS OF LEAST-SQUARES SOLUTIONS

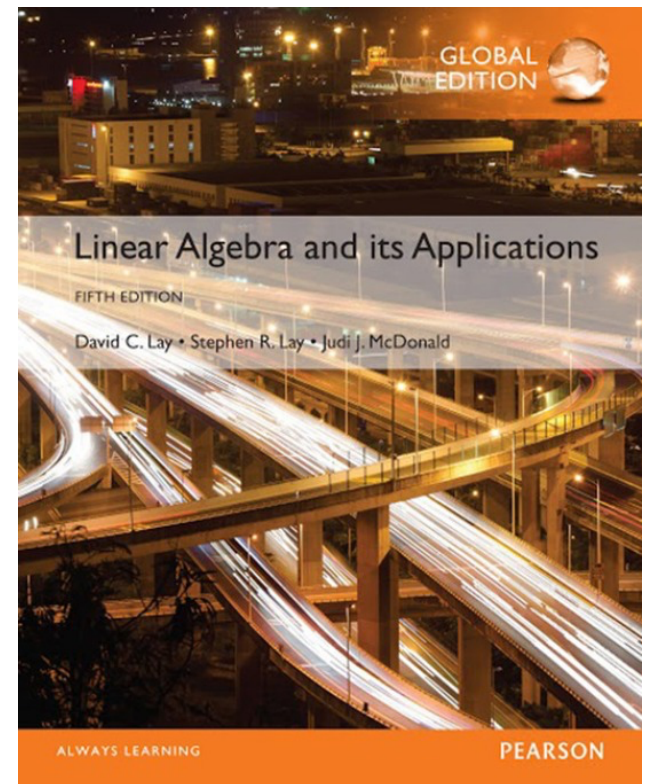
- The columns of Q form an orthonormal basis for $\text{Col } A$.
- Hence, by Theorem 10, $QQ^T\mathbf{b}$ is the orthogonal projection $\hat{\mathbf{b}}$ of $\mathbf{b}$ onto $\text{Col } A$.
- Then $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$, which shows that $\hat{\mathbf{x}}$ is a least-squares solution of $A\mathbf{x} = \mathbf{b}$.
- The uniqueness of $\hat{\mathbf{x}}$ follows from Theorem 14.

6

Orthogonality and Least Squares

6.7

THE GRAM-SCHMIDT PROCESS



INNER PRODUCT SPACES

- **Definition** An **inner product** on a vector space V is a function that, to each pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in V , associates a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ and satisfies the following axioms, for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V and all scalars c :
 1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
 2. $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
 3. $\langle c\mathbf{u}, \mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$
 4. $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ if and only if $\mathbf{u} = \mathbf{0}$
- A vector space with an inner product is called an **inner product space**.

INNER PRODUCT SPACES

- **Example 1** Fix any two positive numbers—say, 4 and 5—and for vectors $u = (u_1, u_2)$ and $v = (v_1, v_2)$ in $\mathbb{R}^2$, set

$$\langle u, v \rangle = 4u_1v_1 + 5u_2v_2 \quad (1)$$

- Show that equation (1) defines an inner product.
- **Solution** Certain Axiom 1 is satisfied, because $\langle u, v \rangle = 4u_1v_1 + 5u_2v_2 = 4u_1v_1 + 5u_2v_2 = \langle v, u \rangle$.

INNER PRODUCT SPACES

- If $w = (w_1, w_2)$, then

$$\begin{aligned}\langle u + v, w \rangle &= 4(u_1v_1)w_1 + 5(u_2v_2)w_2 \\ &= 4u_1w_1 + 5u_2w_2 + 4v_1w_1 + 5v_2w_2 \\ &= \langle u, w \rangle + \langle v, w \rangle\end{aligned}$$

- This verifies Axiom 2. For Axiom 3, compute

$$\langle cu, v \rangle = 4(cu_1)v_1 + 5(cu_2)v_2 = c(4u_1v_1 + 5u_2v_2) = c\langle u, v \rangle$$

INNER PRODUCT SPACES

- For Axiom 4, note that $\langle u, u \rangle = 4u_1^2 + 5u_2^2 \geq 0$, and $4u_1^2 + 5u_2^2 = 0$ only if $u_1 = u_2 = 0$, that is, if $u = 0$.
- Also, $\langle 0, 0 \rangle = 0$. So (1) defines an inner product on $\mathbb{R}^2$.

LENGTHS, DISTANCES, AND ORTHOGONALITY

- Let V be an inner product space, with the inner product denoted by $\langle u, v \rangle$. Just as in $\mathbb{R}^n$, we define the length, or norm, of a vector v to be the scalar

$$\|v\| = \sqrt{(v, v)}$$

- Equivalently, $\|v\|^2 = \langle v, v \rangle$.
- A **unit vector** is one whose length is 1. The **distance between u and v** is $\|u - v\|$. Vectors u and v are **orthogonal** if $\langle u, v \rangle = 0$.

TWO INEQUALITIES

- Given a vector $\mathbf{v}$ in an inner product space V and given a finite-dimensional subspace W , we may apply the Pythagorean Theorem to the orthogonal decomposition of $\mathbf{v}$ with respect to W and obtain

$$\|\mathbf{v}\|^2 = \|\text{proj}_W \mathbf{v}\|^2 + \|\mathbf{v} - \text{proj}_W \mathbf{v}\|^2$$

- See Fig 2 on the next slide. In particular, this shows that the norm of the projection of $\mathbf{v}$ onto W does not exceed the norm of $\mathbf{v}$ itself. This simple observation leads to the following important inequality.

TWO INEQUALITIES

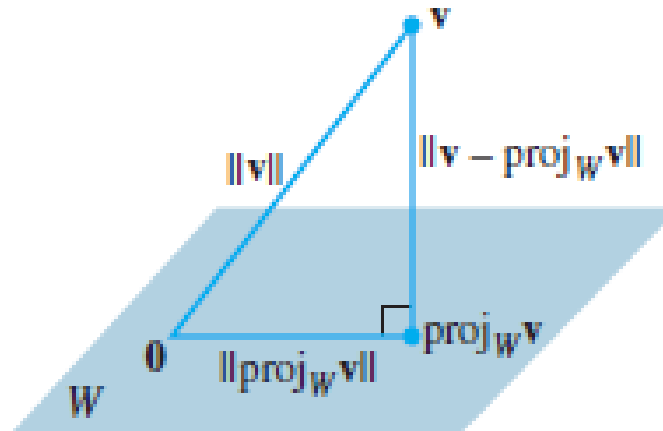


FIGURE 2

The hypotenuse is the longest side.

- **Theorem 16 The Cauchy-Schwarz Inequality:** For all u, v in V ,

$$|(u, v)| \leq \|u\| \|v\| \quad (4)$$

TWO INEQUALITIES

- **Proof** If $u = 0$, then both sides of (4) are zero, and hence the inequality is true in this case.
- If $u \neq 0$, let W be the subspace spanned by u .
- Recall that $\|cu\| = |c| \|u\|$ for any scalar c . Thus
$$\|proj_W v\| = \left\| \frac{\langle v, u \rangle}{\langle u, u \rangle} u \right\| = \frac{|\langle v, u \rangle|}{|\langle u, u \rangle|} \|u\| = \frac{|\langle v, u \rangle|}{\|u\|^2} \|u\| = \frac{|\langle u, v \rangle|}{\|u\|}$$
- Since $\|proj_W v\| \leq \|v\|$, we have $\frac{|\langle u, v \rangle|}{\|u\|} \leq \|v\|$, which gives (4).

TWO INEQUALITIES

- **Theorem 17 The Triangle Inequality:** For all u, v in V ,

$$\|u + v\| \leq \|u\| + \|v\|$$

- **Proof** $\|u + v\|^2 = \langle u + v, u + v \rangle = \langle u, u \rangle + 2\langle u, v \rangle + \langle v, v \rangle$
 $\leq \|u\|^2 + 2|\langle u, v \rangle| + \|v\|^2$
 $\leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2$
 $= (\|u\| + \|v\|)^2$
- The triangle inequality follows immediately by taking square roots of both sides.